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Task Description:	The University of Texas Southwestern Medical Center (UTSW) NSCOR focuses on the development of quantitative dose risk estimates following HZE particle irradiation for the development of key genetic, epigenetic, gene expression, and cellular functional changes in the multi-step pathogenesis of lung cancer in both new human bronchial epithelial cell (HBEC) and transgenic mouse models of lung cancer. The risk of developing these changes are being compared in these models to that of gamma-irradiation. These dose risk assessments are designed to measure events of both radiation-induced promotion (modification of proliferation kinetics of already-initiated cells) as well as radiation-induced initiation (mutational) events. They also measure effects on development of cancer "stem-like" cells ("cancer initiating cells"). 3D organotypic culture and animal models also allow measurement and risk estimation in both tissue surrogates and in vivo lung tissues. These individual risk estimates can then be combined into a model (such as a two-stage clonal expansion model) for overall risk of developing lung cancer from exposure to galactic cosmic radiation. To achieve these goals, UTSW has assembled a team of scientists who are leaders in the study and translation application of the molecular pathogenesis of lung cancer (Minna, Gazdar, Shay); in radiobiology and DNA repair (Chen, Story); a panel of expert Internal and External Advisors (from other NSCORs and lung cancer research experts); as well as expert consultants and collaborators. This team has developed a novel immortalized HBEC system that can be genetically manipulated, studied in 2D monolayer and 3D organotypic cultures. Using these HBECs we have completed many studies with HZE particle irradiation showing that further progression toward lung cancer can be detected and quantified, that specific expression profiles for HZE irradiation exist, and that HBECs genetically manipulated to progress part of the way toward malignancy are even more sensitive to HZE-induced functional changes. In addition, we are completing experiments testing whether transgenic mouse models of lung cancer after low and high LET exposure increase the early onset or progression of lung cancer in the whole animal in real time. The NSCOR has 4 Projects: 1. Genetic and epigenetic changes in human bronchial epithelial cells following exposure to HZE particle irradiation; 2. Effect of HZE particles on DNA damage-sensing and repair pathways in human lung epithelial and fibroblast cells; 3. Effect of HZE particle irradiation on functional progression of human lung cancer at the cellular and organotypic level; and 4. Effects of HZE particles on the development of lung cancer in vivo in novel mouse models. These projects are supported by 4 Cores: Administrative; Cell Culture; Genomics and Proteomics; and Biostatistics and Bioinformatics. In addition, with funding from the Department of Energy (DOE) contributing to this NSCOR, studies of low-LET and low dose radiation in the same model systems are also being undertaken. The HZE particle irradiation is done at Brookhaven National Lab, and the biostatistical analysis shows the experiments are powered to help achieve NASA mandated risk estimate confidence levels. Finally, since this group has substantial UTSW institutional commitments, holds a NCI Special Program of Research Excellence (SPORE) in Lung Cancer, a Center for the NCI Early Detection Research Network (EDRN) program, and is part of the NCI sponsored Genetic Epidemiology of Lung Cancer Consortium (GELCC) there is great synergism and additional resources available for the successful completion of this proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Lung cancer is the leading cause of cancer death in both women and men in the USA and in the Western world. While lung cancer is caused mostly by smoking exposure, a significant number of lung cancers arise in life time never smokers and thus is a sizable health problem in the USA. Indeed lung cancer in never smokers is the 7th leading cause of world-wide cancer related deaths. Likewise, 50% of all new lung cancer cases occur in former smokers (quitting over 5 years before). In all of these scenarios a major underlying question has been the role of other environmental carcinogens especially environmental radiation. One source of this is radon gas in the environment including radon gas exposure in homes with alpha particles (which represent high-LET radiation). Thus, the interaction of radiation in the environment on Earth including from alpha particles in the genesis of lung cancer and quantitation of the effect of such radiation on lung epithelial cells is of major health importance. In addition, in the past decade in the USA, Europe, and Asia there has been a resurgent interest in radiographic screening for lung cancer, particularly using serial computed tomography (CT) scans. In fact the USA NCI has a large (50,000 person) National Lung Cancer Screening Trial (NLST) that randomized persons to have CT scans or not for the early detection of lung cancer. While there is general agreement that CT scans can detect lung lesions better than conventional chest x-rays, and that CT scans do detect many early stage lung cancers, there are a variety of questions concerning the use of this approach because of its cost and substantial radiation exposure. Greater than 25% of persons screened (current or former smokers) will have some lesions requiring further CT scan follow-up, when biopsies are performed only 1% of the overall group will have histologic evidence of lung cancer. In addition, many of the persons will require several CT scans in follow-up to determine if the lesion is growing to warrant a biopsy. Finally, there have been several analyses that suggest that even in CT detected and resected early stage lung cancers it is not clear that such an approach is actually changing lung cancer mortality in the population screen – the gold standard for proceeding with such an approach as a national health policy. Thus, there is considerable interest in knowing the contribution of CT scan doses of gamma radiation to the pathogenesis of lung cancer. This is particularly true since this radiation is being administered to individuals (current and former smokers) who are at highest risk of having preneoplastic lesions leading to lung cancer. Our studies of low dose radiation of the preclinical human and mouse model systems will allow us to provide some risk assessment for the effect of such radiation doses (low dose) on the genesis of lung cancer and provides the rationale for DOE support of our studies. Radiation is a known carcinogenic influence but the molecular events in lung epithelial cells in response to radiation need to be determined. A key element of this is the impact of radiation on lung epithelial cells with preneoplastic lesions. Our NSCOR uses innovative technology of human bronchial epithelial cells (HBECs) developed by us (coming from 45 different individuals) which have also been engineered to have known oncogenic preneoplastic changes (such as expression of oncogenic K-RASv12, knockdown of expression of mutant p53, expression of oncogenic mutant EGFR, expression of c-myc). Our studies are providing quantitative data following high-LET and low dose gamma irradiation on these HBECs. The primary benefits from this work will be the development of quantitative risk assessment models for astronauts of high-LET radiation in space in order to reduce the uncertainties in prediction of risk of astronauts developing lung

cancer as part of long-term missions in space. In addition, as stated we will be developing risk estimates following low dose gamma radiation as seen on Earth, particularly from CT scan studies. Thus, the results will also have major impact and benefit to life on Earth includes new knowledge of the effects on human lung epithelial cells of irradiation in terms of quantitative genetic and epigenetic changes and gene expression changes following irradiation which are of potential importance as markers of radiation exposure that could occur through CT scan screening studies, and environmental exposures, accidentally or through terrorism. A part of this will be to understand the quantitative response of DNA repair pathways to radiation including specific DNA repair components such as DNA repair enzymes and signaling pathways in lung epithelial cells. We are also determining quantitative changes in biologic phenotypes of lung epithelial cells in response to radiation which include colony formation in liquid and soft agar (anchorage independent growth) and differentiation and invasion in 3 dimensional (3D) cultures, and finally growth in vivo as either xenografts (for human cells) or endogenous lung cancers (for mouse models of lung cancer). From all of the above information we will identify novel molecular biomarkers (mRNA, DNA, protein) for lung carcinogenesis. Finally, we are also learning about inter individual variation of the response of HBECs to radiation with respect to the various genetic, epigenetic, mRNA and protein expression, and signaling pathway changes. Our large HBEC panel coupled with new high density single nucleotide polymorphism (SNP) analysis will allow describing polymorphic differences in such responses. Our lung cancer NSCOR is addressing the following specific critical path roadmap and gap questions: "We are improving the understanding of space radiation in lung cancer initiation, promotion and progression spanning cell and molecular biology at the cell, tissue and system biology level that will lead to significant reduction in the uncertainties in risk projection models ; "We are investigating the increased risk from space radiation as a function of age of exposure, age, latency, gender, tissue, radiation quality and dose rate? ; "We are developing tissue specific risk models using human 3D culture and animal models for lung cancer ; "We are investigating how aberrant DNA damage processing, genomic instability, epigenetic effects including methylation, persistent oxidative damage, altered senescence, and non-targeted effects contribute to irradiation-associated lung cancer using a variety of surrogate endpoints that are being validated. Overall we are conducting research to investigate how biological endpoint and systems biology approaches can be used to integrate research on the cellular and molecular as well as tissue level of radiation damage that will lead to significant reductions in the uncertainties in risk projection models of lung cancer. As will be described in this report, we are confident and will provide compelling evidence that the sum of the efforts of our assembled team are significantly contributing more to assessing cancer risk associated with space radiation compared to the individual components.

Project 1. Genetic and epigenetic changes in human bronchial epithelial cells following exposure to HZE particle irradiation. Michael Story, Ph.D. Lead, Adi F. Gazdar, M.D. Co-Lead, John D. Minna, M.D. Co-Lead.
Specific Aim 1: To quantitate the effects of HZE particle irradiation in causing genetic changes in HBECs and their isogenic derivatives.

Specific Aim 2. To quantitate the amount of epigenetic changes caused by HZE particle irradiation in HBECs and their isogenic derivatives using high throughput quantitative methylation specific PCR (MSP) assays for acquired promoter methylation

Specific Aim 3. To quantitate the effects of HZE particle irradiation in causing expression profile (mRNA and protein) changes in human bronchial epithelial cells (HBECs) and their isogenic derivatives containing various lung cancer-related mutations using array based genome wide and multi protein detecting approaches.

Specific Aim 4. To determine the effect of inter-individual variation on the effects of HZE particle irradiation in causing genetic changes, epigenetic changes, mRNA and protein expression profile changes in human bronchial epithelial cells (HBECs).

Project 1: In the last year this project has focused on 3 specific areas. mRNA Expression Analysis. Comparative gene (mRNA) expression analysis using genome wide array technology at equi-toxic doses of low LET gamma radiation vs. that of HZE radiation (28Si and 56Fe) at different time points. We find that there are different mRNA expression profiles generated by each of the different types of radiation and that this is also dependent on the oncogenotype of the HBECs radiated. DNA Methylation Profiles. While there are no global methylation changes we are exploring genome wide changes using array technology as well as quantitative methylation specific PCR (MSP). miRNA Expression Profiling. We are using genome wide array technology to profile miRNA changes after low and high LET radiation and low dose gamma radiation. Project 1 is now close to hitting the milestones described in the original application. We will very shortly have a set of well characterized genomic responses (both mRNA, DNA copy number, DNA methylation, and miRNA expression) to low LET and HZE radiation (Si, Fe) exposures in a series of isogenic immortalized human bronchial epithelial cells (HBECs) which differ only by specific oncogenic changes introduced (e.g. KRAS, p53, EGFR, E6, E7). These variants thus include cells with altered p53, RAS and EGFR signaling. Our goal is to mine the data available in order to develop biomarkers of response that may be used for relative risk determinations of lung cancer from space radiation exposure. Determining such biomarkers dovetails with the transformation studies of Project 3. With the Biostatistics/Bioinformatics Core (Core D) e are using multiple analysis methodologies to develop biomarkers including Gene Set Enrichment, pathway analysis, smar, and Bayesian analysis for interaction nodes. These biomarkers are also being evaluated in mouse models in Project 4 and can be used to address inter-individual differences in carcinogenic response for risk determination.

Project 2: Effect of HZE particles on DNA damage-sensing and repair pathways in human lung epithelial and fibroblast cells. David Chen, Ph.D. Project Lead, Sandeep Burma, Ph.D. Co-Lead.

Specific Objectives: Aim 1. To test the hypothesis that the immediate cellular response to DNA damage induced by HZE would be different and more complex than that induced by X-rays.

Aim 2. To test the hypothesis that persistent DNA damage induced by HZE is more complex and difficult to repair than that induced by X-rays and, therefore, more deleterious.

Aim 3. To determine if damage induced by HZE is preferentially repaired by non homologous end joining (NHEJ) or homologous recombination (HR) or is preferentially repaired in certain phases of the cell cycle.

Aim 4. To compare the response of human bronchial epithelial cells (HBECs) as non immortalized, immortalized, and with premalignant oncogenic changes in 3D cultures to HZE exposure using a high throughput approach.

Project 2: This project has had two major developments this year. First development of human lung epithelial cells with

Task Progress:

endogenous EGFP labeled 53BP1 to allow live cell imaging of DNA repair after low and high LET Radiation in both 2 dimensional (2D) and 3 dimensional (3D). To investigate the cellular responses to DNA double strand breaks (DSBs) induced by low- and high-LET irradiation (IR) in vivo, we generated a human bronchial epithelial cell (HBEC3) line that stably expresses near-physiological levels of EGFP-53BP1 and used this human lung epithelial cell line in a three-dimensional (3D) model tissue culture system (Matrigel) to study induction and repair of DNA DSBs under conditions that were close to those in 3D tissues in vivo by monitoring the formation of 3BP1 foci development and resolution with by monitoring the EGFP labeled endogenous protein. We found that survival curves were identical for HBECs with and without the 53BP1 expression construct. Before we used this system for DSB repair study, we extensively characterized these cells after growing them in Matrigel. These analyses indicated that when we grow HBECs in this extracellular matrix (ECM) culture, they attain a growth-arrested state at day 6, similar to 3D cultures of mammary epithelial cells.

Second, use of HEBECs with endogenous EGFP-53BP1 to study foci dissolution kinetics data after different types of radiation. These studies indicated that a significant proportion of iron-ion-induced damage was irreparable both in 2D and 3D structures, but the number of persistent foci was significantly higher in 3D structures. Collectively, these EGFP-53BP1 foci dissolution analyses suggest that the DNA DSBs induced by high-LET IR are differentially repaired in 2D and 3D structures.

Project 3. Effect of HZE Particle Irradiation on Functional Progression of Human Lung Cancer. John D. Minna, M.D. Project Lead, Jerry Shay, Ph.D. Co-Lead.

Specific Aim 1. Effects of HZE particles on early events in lung cancer progression (soft agar colony formation and other markers of cellular transformation).

Specific Aim 2. Effects of HZE particles on intermediate events in lung cancer progression (invasion in 3 D organotypic cultures).

Specific Aim 3. Effects of HZE particles on late events in lung cancer progression (ability to form tumors in immune deprived mice).

Project 3: We have made substantial progress in the determination of transformation by HZE particles in normal human bronchial epithelial cells (HBEC3KT). (Transformation as defined by the ability to grow in soft agar.) We last reported that there was a peak of transformation (two logs higher from a base rate of 10-7 to ~10-5) for 56Fe particle irradiations at 0.25 Gy which then diminished with higher doses and was more marked than the increase see with gamma radiation at 2-4 Gy. These data are updated in with the addition of 28Si data, which also shows the same response. Interestingly, the HBEC3KT Kras/p53 cell line (which has a four log higher spontaneous transformation rate (10-3) did not show any increase in transformation rate by gamma, Si, or Fe up to 1 Gy. We have now isolated over 160 individual transformed foci and these show that ~50% of the clones breed true for this phenotype. We have studied these and found that they exhibit epithelial to mesenchymal transition (EMT) compared to non irradiated control transformed foci by both morphology and molecular markers (vimentin, ecadherin expression). In addition, they have an altered radiation survival curve phenotype. With these cell isolates in hand we will begin to determine tumorigenicity. Tumors themselves will be examined for radio sensitivity and aggressiveness. Tumor phenotype will be compared to those mouse lung carcinomas generated after Fe-exposure in Project 4, which were characterized as especially aggressive. Those cells that develop tumors will be characterized at the genomic and proteomic level and compared against the phenotype of human lung tumors generated via smoking or tumors seen in non-smokers, or via radiation-exposure, in particular via inhalation of radioisotopes such as Pu or Rn. This should allow the characterization and risk for radiation-induced lung cancers, particularly those induced by HZE exposure.

Project 4. Effects of HZE Particles on the Development of Lung Cancer in Novel Mouse Models. Jerry W. Shay, Ph.D. Project Lead, James Richardson, DVM, Ph.D. Co-Investigator, Oliver Delgado, Student Assistant.

Specific Aim 1. Use the K-RAS transgenic mouse model of human lung cancer to determine effects of low- and high-LET radiation in tumor initiation and progression.

Specific Aim 2. Use additional transgenic mouse models to determine the effects of low- and high-LET radiation in both living animals and in isolated tissues.

Summary of Project 4: 1. Fractionated and single dose irradiation of 1Gy 56Fe- particles do not increase tumor incidence

2. Tumors in the LA1 K-RAS mice irradiated with fractionated doses of (0.2Gy/day, total 1 Gy (1GeV/n) 56Fe-particles have a statistically significant increased probability of progressing to invasive carcinoma in comparison to a single acute 1Gy 56Fe- dose.

3. These results suggest significant effects of fractionated high-LET irradiation on cancer progression

4. To test this hypothesis we have initiated additional single dose and fractionated doses to confirm these important observations. We have irradiated mice with fractionated x-ray (0.6 Gy/day x 5 days) and have not observed an increase incidence of invasive tumors. We have initiated fractionated 56Fe- particle experiments (0.1 Gy/day x 10 over a 12 day period).

5. We are testing the hypothesis that fractionated high-LET irradiation induces persistent inflammatory signaling that does not occur in single doses. An alternative hypothesis is that there could be a role of stem cell repopulation for lung tissue that is important in cancer progression. Thus, acute doses of high-LET ions may increase cell killing, reducing the number of target stem cells available for initiation and progression to lung cancer.

6. We have provided non lung tissues from our control and irradiated mice to other investigators (Ameila Eisch, brain tissue; Zhi-Min Yuan, kidney, liver, spleen)

7. Mouse telomerase knockout model of lung cancer. Our rationale and overall goal is to produce a mouse with “humanized” telomere length so that we can irradiate with X-ray, proton and HZE particles to determine genomic changes and DNA damage/repair in the lung and other tissues. Inbred strains of mice have extraordinarily long telomeres and thus much of the genomic instability due to telomeres shortening in humans is unlikely to be observed in typical mouse irradiation studies. We have reached the third generation of breeding mTERT KO to KO to obtain short (more human like) telomere length and initiated our first experiments in the Fall 2009 NSRL run. These animals have

been returned from the BNL and we will be conducting pilot microarray experiments to compare to similarly irradiated wild-type mice for DNA damage and inflammatory changes that may distinguish mice with humanized telomeres from typical inbred strains of mice. Future experiments will be initiated to determine if more end joining, end-end fusions, cytogenetic alterations, micronuclei, ongoing reactive oxygen species (ROS) occur when these mice are exposed to low- and high-LET irradiation.

8. EGFR tyrosine kinase domain mutation mouse/ Our rationale for initiating experiment with a new mouse model is that astronauts do not smoke. EGFR mutations are frequently found in lung cancers from never smokers. The models we have chosen to use have an EGFR mutation or small deletion that can be induced by adding doxycycline in the food or drinking water. We now have the mice in our breeding colony and have been scaling up breeding pairs. Experiments under consideration when sufficient numbers of animals are available include the following: We will activate the EGFR-mut about a month prior to low- and high-LET irradiation to compare DNA damage and repair in tumor versus normal tissue; We will also subject these animals to low- and high-LET irradiation prior to activating the EGFR mutation and then add doxycycline to the food for three months. These experiments are designed to further dissect stromal/epithelial differences in cancer progression. The hypothesis that we propose to test is that by activating the stromal compartment prior to tumor initiation the host environment may be permissive for the future growth of the tumors; We will compare the EGFR-mut inducible model to the inducible K-RAS mouse model.

9. DOE Related studies: One aspects of our NSCOR is supported by the DOE. While low doses or even low dose rates have not yielded informative animals results in most instances, we have now formulated a series of testable hypothesis that may be tenable. We propose to send both the mutant EGFR and K-RAS mice to the Colorado State University (CSU) low dose facility and then expose them to low dose rates (10 cGy/day) for 10, 20 and 30 days then activate mutant EGFR or K-RAS. We will be able to address the central and unanswered question if tumors appear faster upon activation of the oncogene after low dose exposure to the stroma). We can also determine if activated tumor will or will not regress when the doxycycline food is removed at 90 days.

Core A: Administrative Core. Leads: John Minna and Jerry Shay, and UTSW NSCOR Administrator: Brenda Zielke.

The Administrative Core has established accounts and control over disbursement of all funds from this NSCOR application. It has also kept track of all personnel employed by this NSCOR. It has provided a central place for communication and coordinated data sharing In addition, it has established a series of biweekly to monthly scientific and planning meeting of the NSCOR investigators at UTSW. Approximately 25 scientific personnel attend each of these meetings. At each meeting administrative matters are discussed, experiments are planned, administrative changes made by the senior leadership team, and most importantly a scientific presentation by one of the NSCOR investigators is made and critiqued by the other investigators. Dr. Minna serves as the Administrative Core Lead with Dr. Jerry Shay as the Administrative Core Co-Lead. The Administrative Core coordinated all of the arrangements for the participation of several NSCOR investigators for the NASA reverse site visit February 18-19, 2009, and travel for several investigators to the 20th NASA investigators meeting in Cologne Germany. The Administrative core also arranged the Internal Scientific Advisory Board Review (March 4, 2008 as part of a mini-retreat at UT Southwestern) and the External Advisory Board Review (October 15, 2008 Mary Helen Barcellos-Hoff, Robert Ullrich and Martin Brown participating). Finally, the Administrative Core coordinated travel of multiple personnel to BNL for experimental work.

Core B. Cell Culture. Core Leads: John D. Minna, M.D. and Jerry Shay, Ph.D.

Enhancement of human bronchial epithelial cell 3D lung model of tissue like development. There is increasing evidence that cancer arises from stem cell populations that are important for the normal turnover of tissues and the rapid repair of tissues in instances of injury. In the lung there is evidence for a bronchiole alveolar stem cell (BASC) population that may give rise to lung carcinoma. We developed a panel of immortalized human bronchial epithelial cells and have previously demonstrated that these cells when placed in an air medium interface overlying a fibroblast stromal collagen matrix could differentiate into mucous and ciliated bronchial epithelial cells. These results were expected since the cells originated from the central airways and were obtained from mid size bronchial tissues. We examined these cells for a variety of differentiated cell markers and observed that they expressed p63 (a transit amplifying cell marker) and surprisingly surfactant protein A (SP-A), CC10, a Clara cell marker, and when differentiated weak expression of the thyroid transcription factor TITF1 (known to be important in adenocarcinomas of the peripheral airway of the lungs). These peripheral markers associated with Clara and type II alveolar cells imply that our immortalized HBECs have enhanced plasticity and should be considered pluripotent lung stem cells. We next placed these cells into Matrigel overlying a lung fibroblast (IMR90) feeder layer and observed that the cells could efficiently differentiate into spheroids or cyst like structures. Studies of these HBECs growing in Matrigel and forming cysts and SACs indicated they started to express SP-A and morphologically resembled small bronchioles and alveolar sacs. When examined in the electron microscope, these cells also had lamellar bodies which are a hallmark of type II alveolar cells. This new improved 3D model of lung cell differentiation has many advantages over the previous model. First, they can differentiate in 3-5 days in vitro making it feasible to set these up at BNL for future NSRL runs. This is in contrast to the air liquid interface for differentiation into ciliated and mucous secreting bronchial cells which require a minimum of two weeks of culture and our previous experience indicated that they do not ship well to BNL. Second, these differentiating small airways cells resemble the type of lung cancer that is likely to be increased in astronauts on long-term space missions. Thus, during the 2008 and 2009 NSRL runs we initiated our irradiation experiments with these cells (see Project 2 overview). Importantly, we introduced a reporter gene 53BP1 containing EGFP so that they sites of irradiation induced damage could be visualized in real time. One final improvement in the 3D culture of our immortalized HBECs consists of culturing them on top of Matrigel. Under these conditions, the cells form web like structures then bud as if they are formed alveoli. This new method also permits a way to have 3D cultures in 5 days and thus promises to be a new addition to our arsenal of ways to examine the influence of irradiation on lung cells under various conditions.

Core C. Genomics and Proteomics Core. Core Lead: Michael Story, Ph.D., Ching-Reng Yang, Ph.D. Core Co-Lead.

Genomics Core: The Core has continued to perform genome wide profiling including mRNA, miRNA, DNA SNP copy number, and DNA methylation, as well as reverse phase protein array (RPPA) profiling on samples before and after radiation in support of all of the projects of this NSCOR. In addition, this Core interfaces with the Biostatistics and Bioinformatics/Database Core (Core D) to ensure the primary results from these genome wide studies are available for appropriate biostatistical and bioinformatics analysis. These are very precise technologies that requires extensive experience to perform and sophisticated instrumentation to carry out. Furthermore, having a single facility through which all array analysis passes minimizes the variability brought about by different individuals processing samples even though the same platform is being used. For these reasons we included a Core facility (Core C, Genomics and

Proteomics) in this NSCOR grant. This NSCOR Core also interfaces with the UTSW Cancer Center Microarray and the UTSW school wide Cores.

The NSCOR Core has the following duties: 1. Interact with Project Leaders and investigators in the design of experiments that use microarray technologies (mRNA, miRNA, DNA copy number, and proteomics) in their experimental approach. Biostatistics and bioinformatics will also be integrated in this process. 2. Establish procedures for consistency in sample preparation and RNA, DNA, and protein extraction, test all RNA, DNA, and protein samples for integrity, and provide reports to that effect. 3. Perform all hybridizations, perform all scans, and place data on a secure server. 4. Provide early analysis on array quality control and preliminary data analysis to investigators. 5. Interact with the Biostatistics and Bioinformatics Core (Core D) to provide bioinformatic analysis of data according to project requirements.

NSCOR Core D: Biostatistics and Bioinformatics. Core Lead: Xian-Jin Xie, Ph.D.; Core Co-Lead: Luc Girard, Ph.D. and Yang Xie, Ph.D.

The Biostatistics and Bioinformatics Core (Core D) provides comprehensive biostatistics and bioinformatics expertise to ensure the statistical integrity and to optimize data analysis of the University of Texas Southwestern (UTSW) NSCOR studies. In doing so, Core D assists investigators with testable hypothesis formulations and proper planning of experiments, manages experimental data generated by NSCOR projects, conducts and advises data analysis, and assists in interpreting final results. To play an integrated role and to serve all the NSCOR projects in a timely manner, Core D biostatisticians Drs. Xian-Jin Xie, Luc Girard, Yang Xie, Chul Ahn and Song Zhang have participated in all monthly project meetings, ensuring that proper biostatistics and data management considerations are adequately taken during all phases of planned experiments. In conjunction with the Genomics (Expression Profiling and Proteomics) Core (Core C) and the Cell Culture Core (Core B), the Biostatistics and Bioinformatics Core (Core D) has the following specific aims:

- To provide database support and expertise for the collection, storage, integration, and retrieval of all sorts of data generated from NSCOR projects.

- To provide valid and optimal statistical design and to conduct proper analysis required to address the specific aims of each project.

- To review background and rationale, to assist in the design, evaluation, and analysis of new research proposal arising from the individual projects.

- To assist the other Cores and Projects in the proper analysis and interpretations of different types of bioassays, particularly mega-data such as those generated from gene expression arrays.

- To assist in manuscript preparation and to review scientific submissions.

- To assist the NSCOR team in building radiation cancer risk models.

As we progress on all the NSCOR projects, significant amount of new data have been generated since our 2008 annual progress report. As a result of Core D effort, a comprehensive data management system has been established that allows efficient access and sharing of all NSCOR data. Data Collection Close to 10,000 samples (including cell pellets and cryogenically preserved cell lines) and experiments (irradiation, microarrays, mouse data) have so far been collected and stored in the NSCOR database. These data can be viewed or edited by any participating investigator provided they have the necessary permissions.

Bibliography Type:	Description: (Last Updated: 11/20/2025)
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Abstracts for Journals and Proceedings	Burong Hu, Asaithamby A, Chen DJ. "Cellular responses to clustered DNA damages induced by high energy and high-Z particles irradiation." Presented at the 55th Annual Radiation Research Society meeting, Savannah, Georgia, October 4-7, 2009. 55th Annual Radiation Research Society meeting, Savannah, Georgia, October 4-7, 2009. , Oct-2009
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